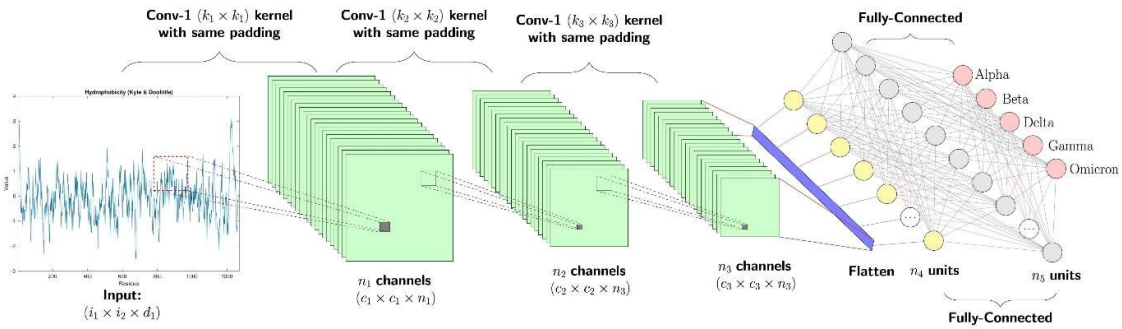
Handwritten Digit Classification (MNIST) Using CNN

1. Rapid Accuracy Improvement

In my notebook, the CNN demonstrated an incredibly fast learning rate. By the end of just the first epoch, the validation accuracy had already reached 98.19%. Over 5 epochs, the training accuracy climbed rapidly to 99.49%, and the final evaluation on the unseen test set yielded a 99.15% accuracy. This rapid convergence to high accuracy happens much faster and peaks higher than a standard ANN.

2. Why CNNs Outperform ANNs for Image Classification



While traditional ANNs can classify images, I found that CNNs are a much better choice for this task due to four main advantages:

- **Preservation of Spatial Structure:** An ANN required me to flatten the 28x28 image into a 1D array of 784 pixels right away, destroying the spatial relationships between pixels. My CNN takes the 2D image as is, allowing it to understand how neighboring pixels connect to form lines and curves.
- **Hierarchical Feature Extraction:** My Conv2D layers act as sliding filters. The early layers learn to detect simple edges, while deeper layers combine those edges into complex patterns (like the cross in a "4"). ANNs only look at raw pixel brightness and miss these geometric shapes.
- **Translation Invariance:** Because my CNN shares filter weights as it scans across the entire image, it can recognize a digit even if it is shifted off-center. An ANN often fails if the object isn't in the same spot, it was during training.
- **Parameter Efficiency:** By utilizing shared filters and MaxPooling2D to shrink the image representations, my CNN model only requires 121,930 parameters. A fully connected ANN of similar depth would need significantly more parameters, making it computationally heavier and more likely to overfit the training data.